香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Measurement-Free Error Correction in Rydberg atom array

Han Wang

HKUST(GZ)

2025-09-16

Outline

Background

Measurement-free error correction

Model

Simulation process

Bibliography

Outline

Background

Measurement-free error correction

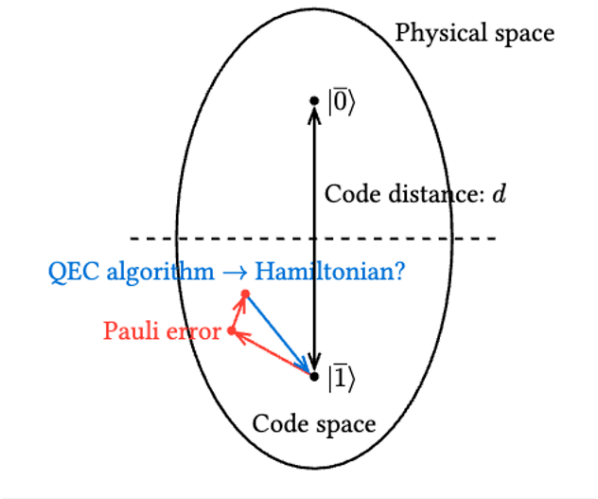
Model

Simulation process

Bibliography

Quantum error correction and code space

- Code space, redundancy, syndrome, error correction.

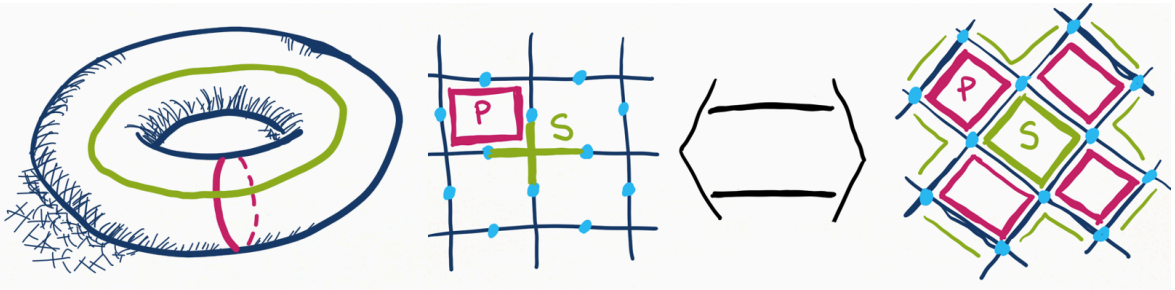


Toric code

- Toric code Hamiltonian:

$$H = -\sum_s S_s - \sum_p P_p$$

where $S_s = \prod_{i \in s} Z_i, P_p = \prod_{j \in p} X_j$



- Stabilizer:
- Stabilized: Eigenvalue of all stabilizers is $+1$.

Main problem

- Measurement and feedback is time-consuming.

(movement-100μs[1], [2]; illuminating and collecting-500μs[3], [4])

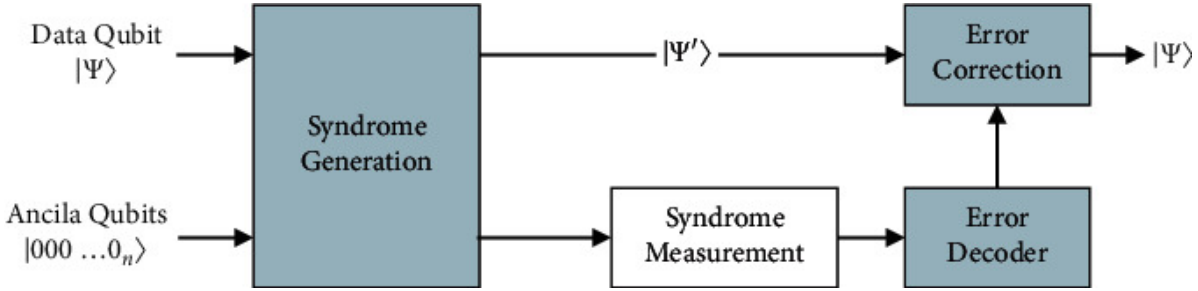


Figure 3: quantum error correction process [5]

- Alternative: self-correcting quantum code(using Hamiltonian to confine to the ground state)[6]

Outline

Background

Measurement-free error correction

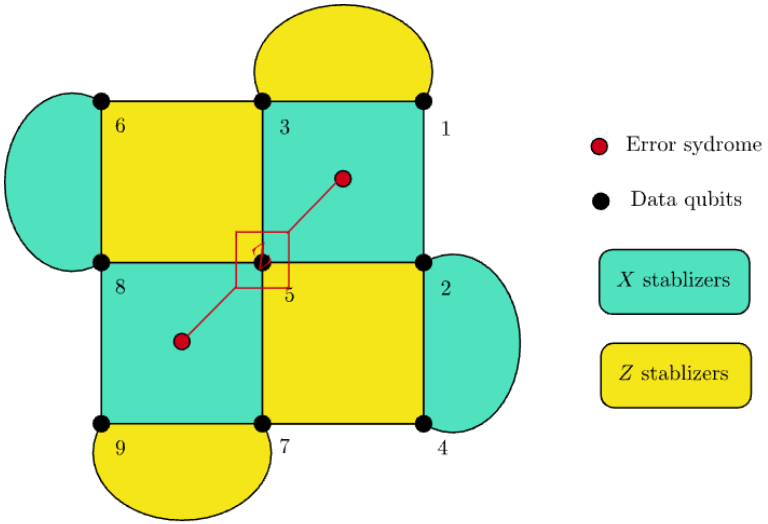
Model

Simulation process

Bibliography

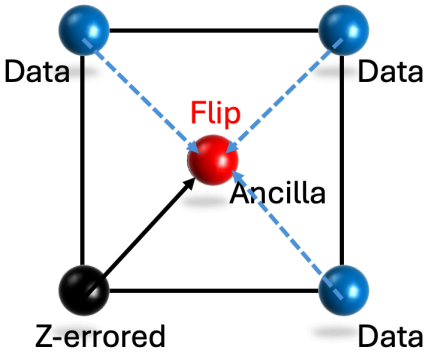
Third approach

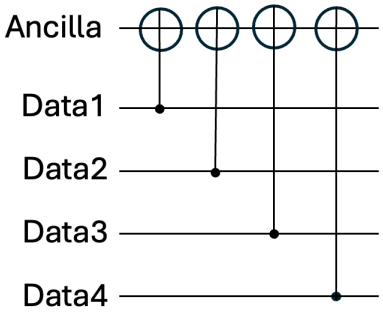
- Measurement-free error correction scheme[7]:



Parity-check process is time consuming[8]

- Moving to entangling zone ⇒ Gate operation ⇒ Back to storage ⇒ Repeating.





- Movement is slow! The reported gate time is about $1\mu s$[9] and the movement time is about hundreds of μs.[1], [2]

Outline

Background

Measurement-free error correction

Model

Simulation process

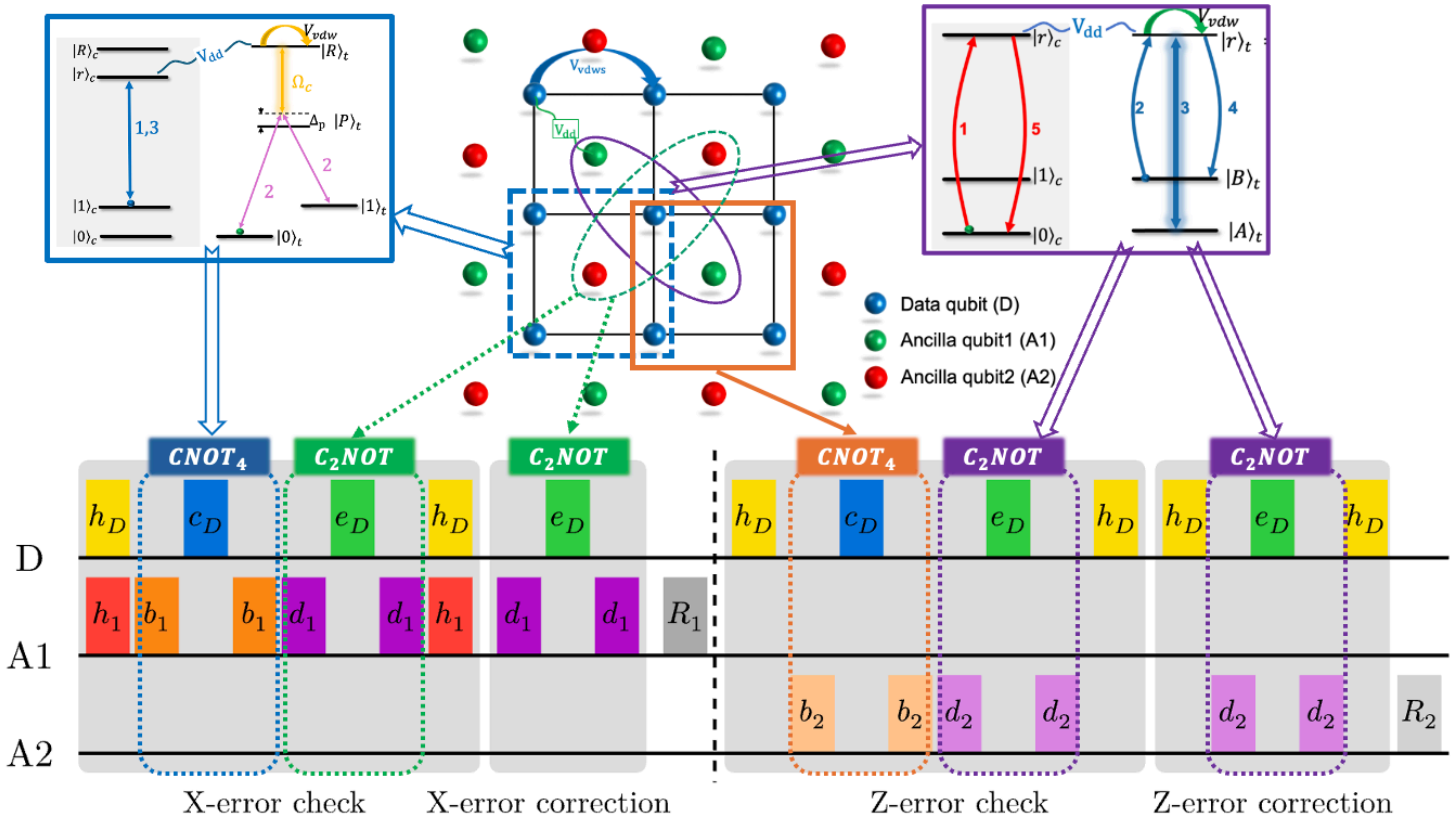
Bibliography

香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Measurement-free and movement-free error correction model

Control qubit in state $|1\rangle \Rightarrow$ target qubit flip: $|0\rangle \Leftrightarrow -|1\rangle$.
Control qubit in state $|0\rangle \Rightarrow$ target qubit keep invariant.

Control qubits in state $|1\rangle \Rightarrow$ target qubit flip: $|0\rangle \Leftrightarrow -|1\rangle$.
One of ontrol qubit in state $|0\rangle \Rightarrow$ target qubit keep invarian

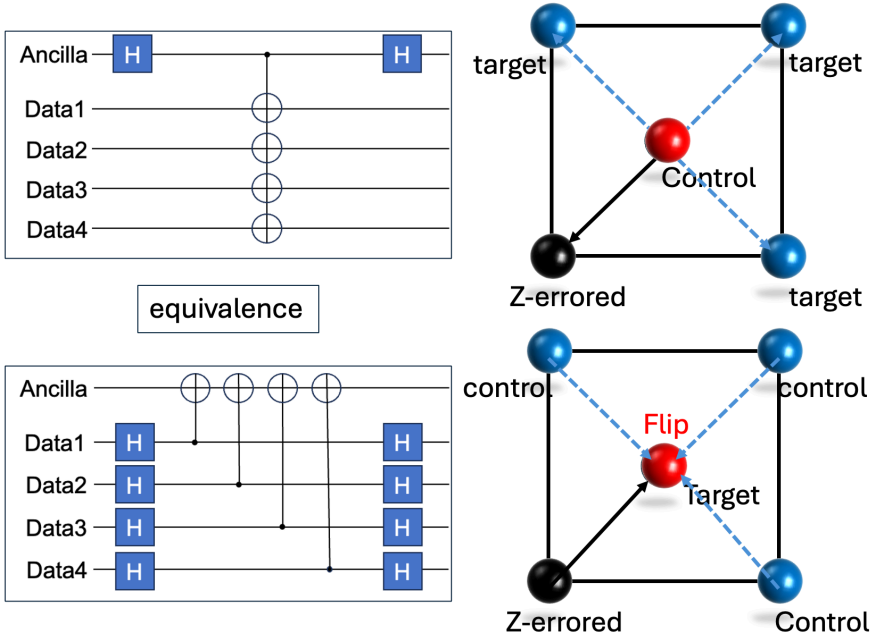


香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Step1: Movement-free parity-check process

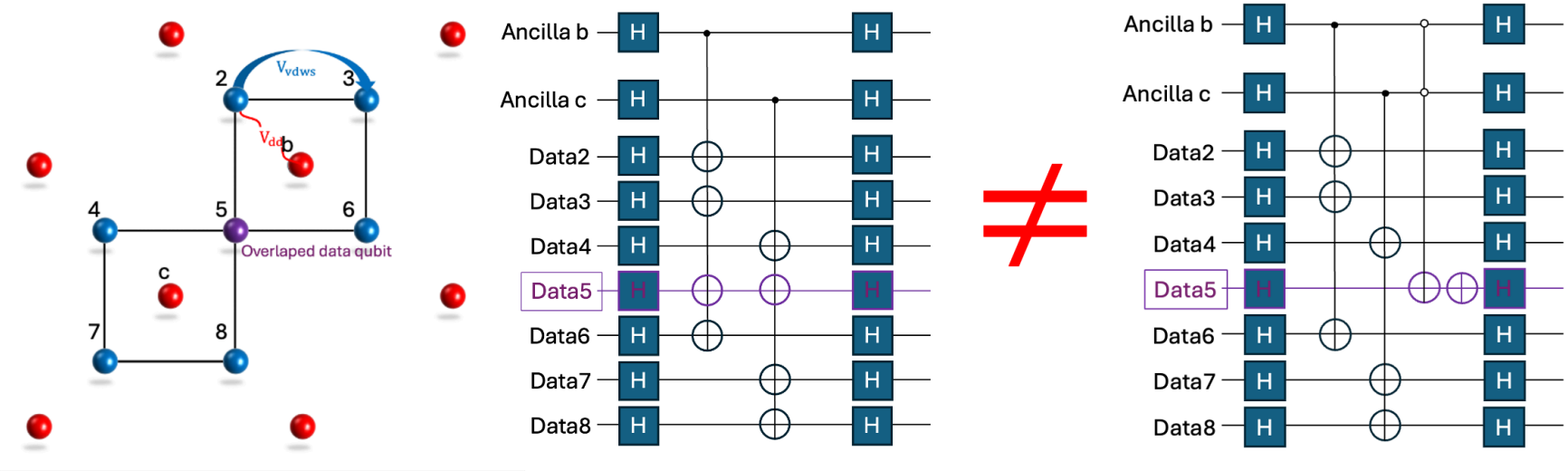
- The one-step parity check process in one plaquette:

– The effective parity check can be implemented with ancilla qubit as control qubit and data qubit as target qubit.



Step1: Movement-free parity-check process

- The one-step parity check process in more than one plaquettes:



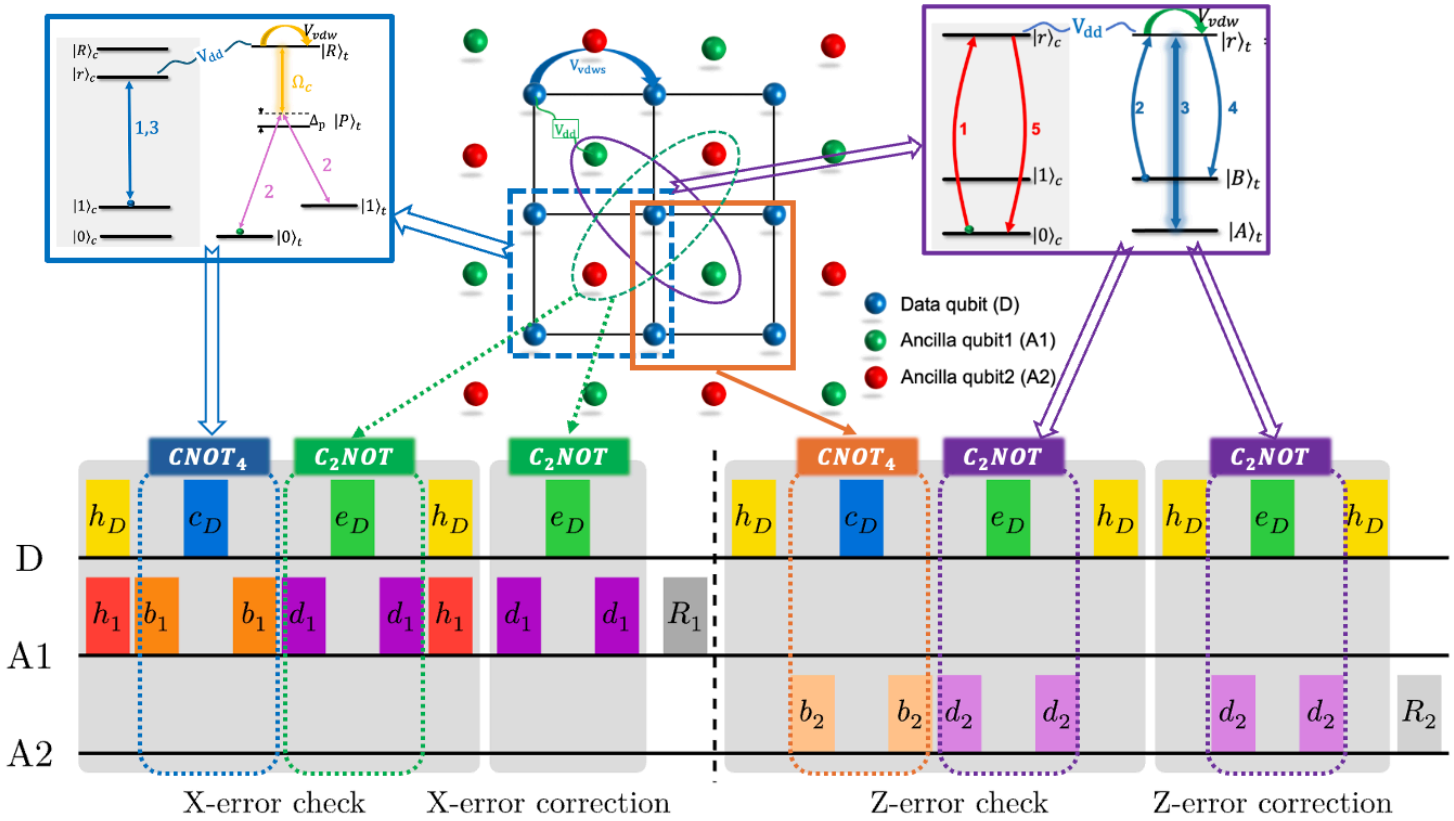
Modifying the pulse sequence to add a CCNOT gate with neighboring ancilla qubit as control qubit and data qubit as target qubit.

香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Measurement-free and movement-free error correction model

Control qubit in state $|1\rangle \Rightarrow$ target qubit flip: $|0\rangle \Leftrightarrow -|1\rangle$.
Control qubit in state $|0\rangle \Rightarrow$ target qubit keep invariant.

Control qubits in state $|1\rangle \Rightarrow$ target qubit flip: $|0\rangle \Leftrightarrow -|1\rangle$.
One of ontrol qubit in state $|0\rangle \Rightarrow$ target qubit keep invarian



Outline

Background

Measurement-free error correction

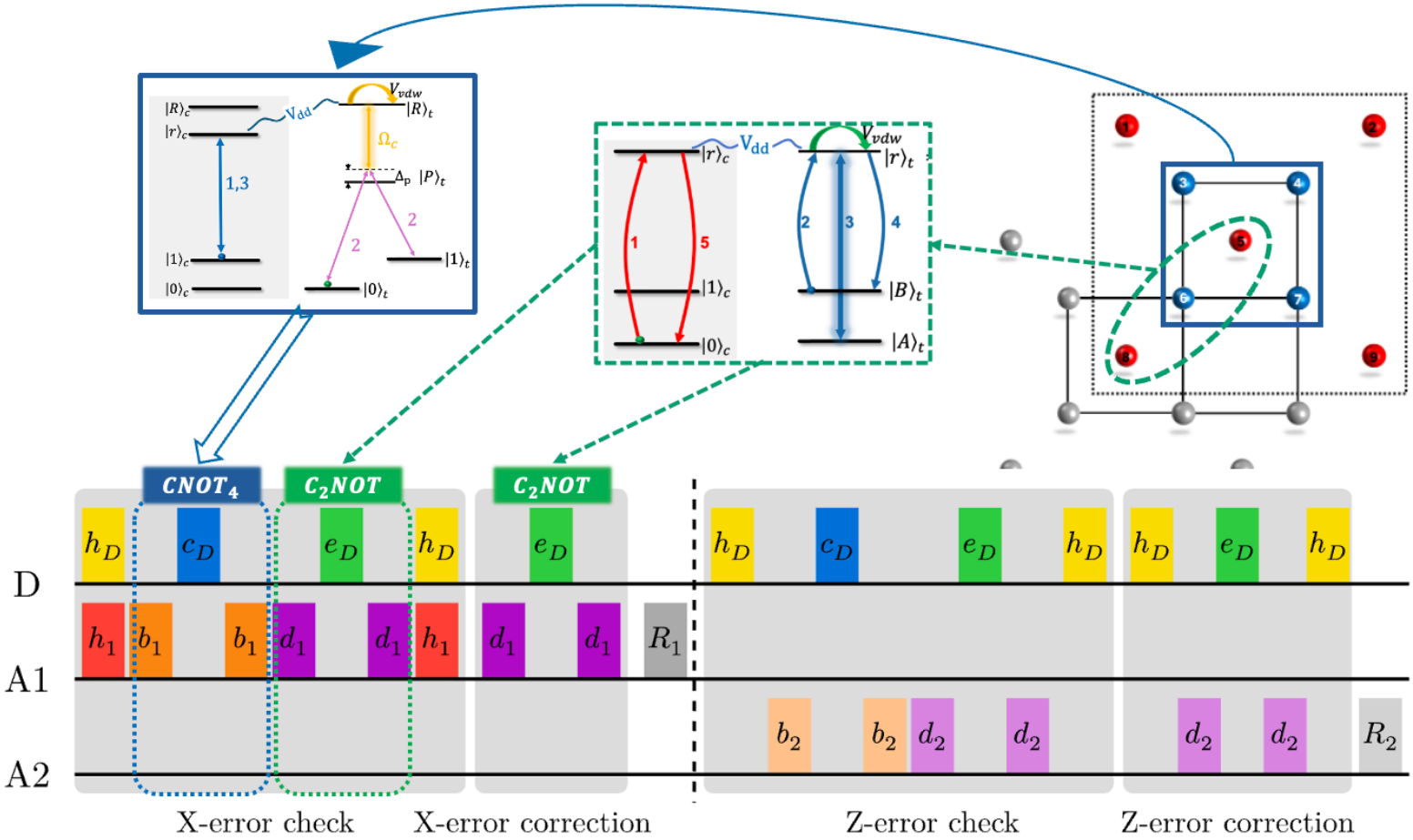
Model

Simulation process

Bibliography

香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

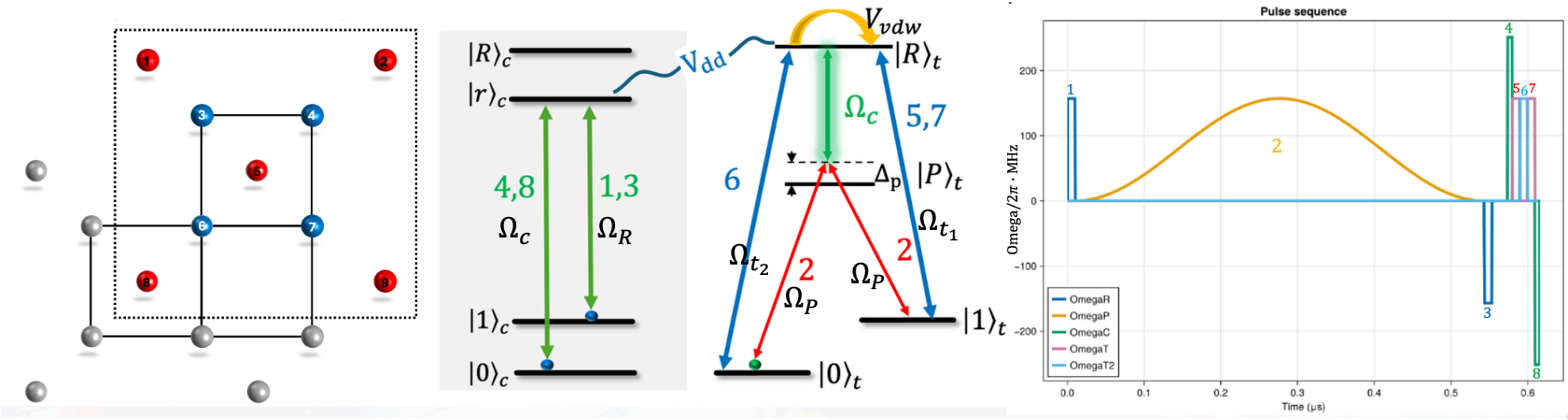
Designed pulse sequence for X-error parity check



香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Designed pulse sequence for X-error parity check

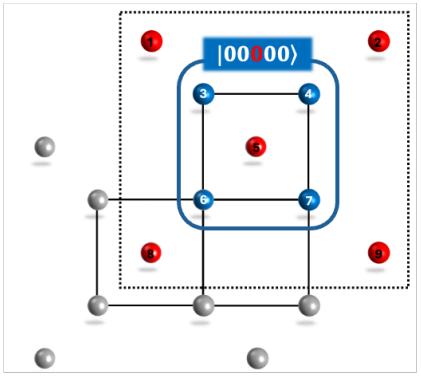
- Including CNOT_4 gate and CCNOT gate:

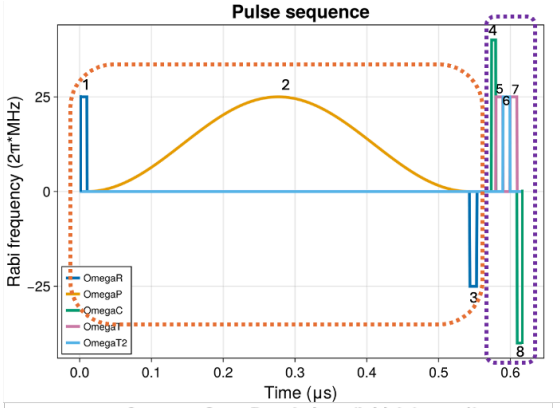


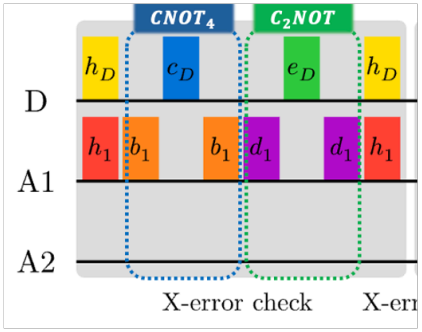
香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

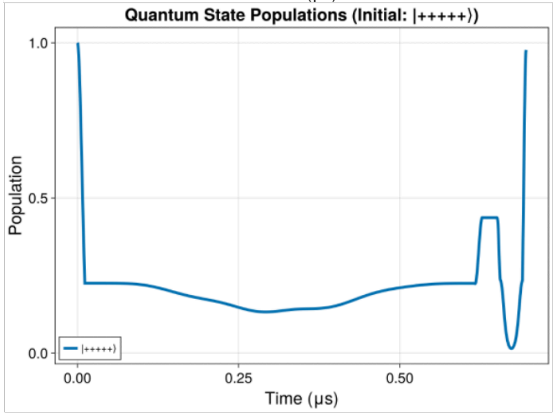
Simulation results for X-error parity check–no error

- $|00000\rangle \Rightarrow |+++++\rangle \Rightarrow$ pulse sequence $\Rightarrow |+++++\rangle \Rightarrow |00000\rangle$(final state)





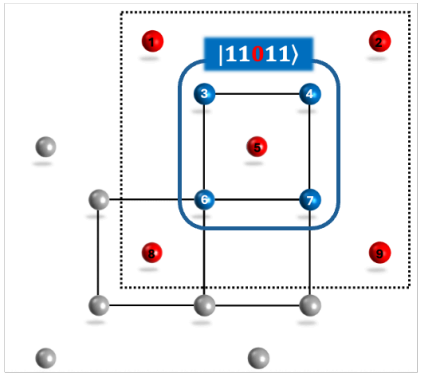


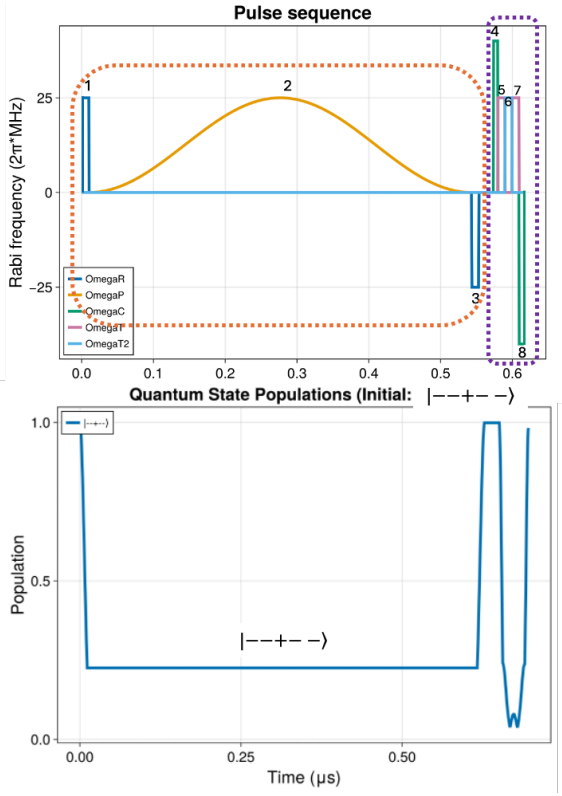


香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Simulation results for X-error parity check–no error

- $|11011\rangle \Rightarrow |--+--\rangle \Rightarrow$ pulse sequence $\Rightarrow |--+--\rangle \Rightarrow |11011\rangle$(final state)

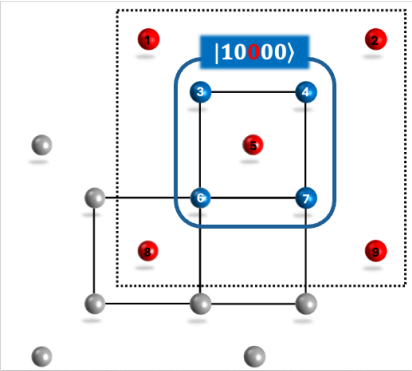


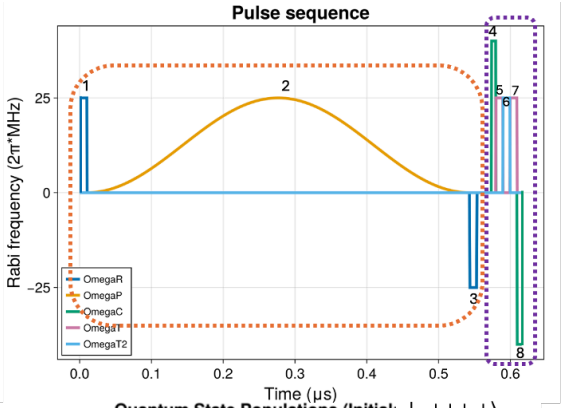


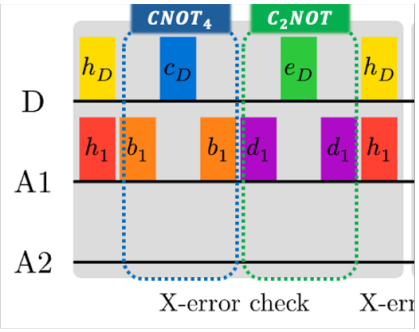
香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

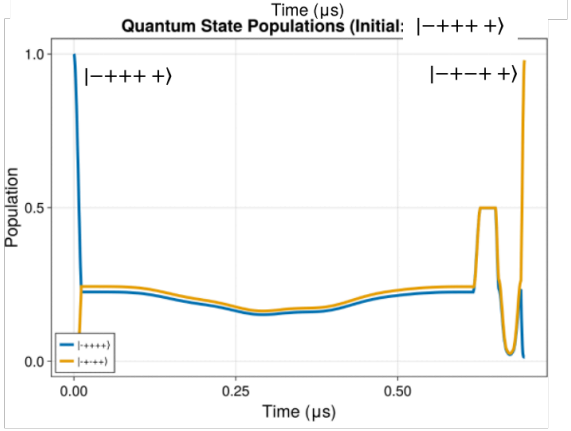
Simulation results for X-error parity check–single error

- $|10000\rangle \Rightarrow |\text{-++++}\rangle \Rightarrow$ pulse sequence $\Rightarrow |\text{-+-++}\rangle \Rightarrow |10100\rangle$(final state)





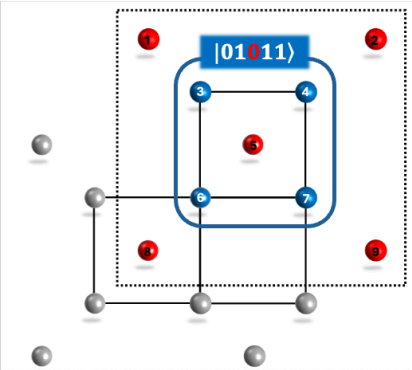


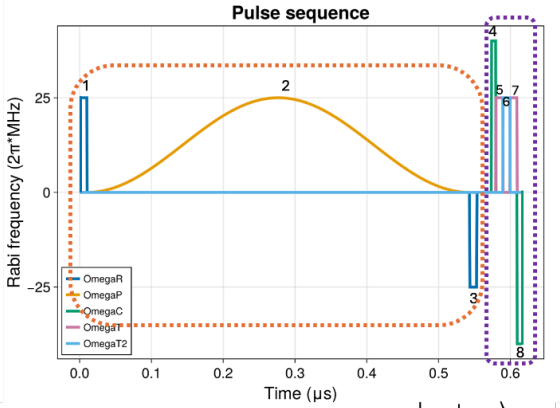


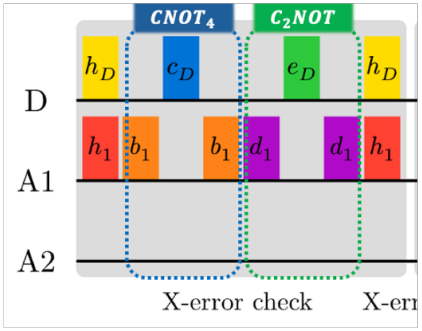
香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

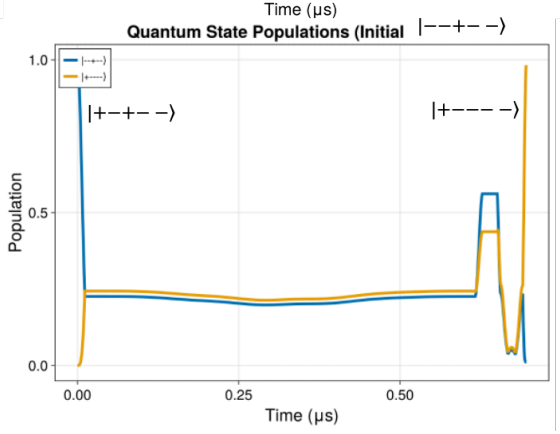
Simulation results for X-error parity check–single error

- $|01011\rangle \Rightarrow |+-+--\rangle \Rightarrow$ pulse sequence $\Rightarrow |+++++\rangle \Rightarrow |00000\rangle$(final state)









Bibliography

Bibliography

[1] D. Bluvstein *et al.*, "A quantum processor based on coherent transport of entangled atom arrays," *Nature*, vol. 604, no. 7906, pp. 451–456, Apr. 2022, doi: 10.1038/s41586-022-04592-6.

[2] D. Bluvstein *et al.*, "Logical quantum processor based on reconfigurable atom arrays," *Nature*, vol. 626, no. 7997, pp. 58–65, Dec. 2023, doi: 10.1038/s41586-023-06927-3.

[3] M. A. Norcia *et al.*, "Midcircuit Qubit Measurement and Rearrangement in a <mml:math xmlns:mml="http://www.w3.org/1998/Math/MathML" display="inline"><mml:mrow><mml:mmultiscripts><mml:mrow><mml:mi>Yb</mml:mi></mml:mrow><mml:mprescripts/><mml:none/><mml:mrow><mml:mn>171</mml:mn></mml:mrow></mml:mmultiscripts></

Bibliography

mml:mrow></mml:math> Atomic Array," *Physical Review X*, vol. 13, no. 4, Nov. 2023, doi: 10.1103/physrevx.13.041034.

[4] T. M. Graham *et al.*, "Midcircuit Measurements on a Single-Species Neutral Alkali Atom Quantum Processor," *Physical Review X*, vol. 13, no. 4, Dec. 2023, doi: 10.1103/physrevx.13.041051.

[5] O. O. Khalifa, N. Amirah Bt Sharif, R. A. Saeed, S. Abdel-Khalek, A. N. Alharbi, and A. A. Alkathiri, "Digital System Design for Quantum Error Correction Codes," *Contrast Media & Molecular Imaging*, vol. 2021, p. 1101911, 2021, doi: 10.1155/2021/1101911.

[6] B. Yoshida, "Feasibility of self-correcting quantum memory and thermal stability of topological order," *Annals of Physics*, vol. 326, no. 10, pp. 2566–2633, Oct. 2011, doi: 10.1016/j.aop.2011.06.001.

Bibliography

[7] S. Heußen, D. F. Locher, and M. Müller, "Measurement-Free Fault-Tolerant Quantum Error Correction in Near-Term Devices," *PRX Quantum*, vol. 5, no. 1, Feb. 2024, doi: 10.1103/prxquantum.5.010333.

[8] D. Barredo, V. Lienhard, S. de Léséleuc, T. Lahaye, and A. Browaeys, "Synthetic three-dimensional atomic structures assembled atom by atom," *Nature*, vol. 561, no. 7721, pp. 79–82, Sep. 2018, doi: 10.1038/s41586-018-0450-2.

[9] H. Levine *et al.*, "High-Fidelity Control and Entanglement of Rydberg-Atom Qubits," *Physical Review Letters*, vol. 121, no. 12, Sep. 2018, doi: 10.1103/physrevlett.121.123603.